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My Learning Experience

In this Lab 07, I worked alone on every task. My contribution is 100 percent. I wrote all code, ran all models, and answered every question.

I want to share my thoughts about this lab. I learned many new things about classification metrics. I looked at one confusion matrix. I saw that this model makes mistakes in 14% of cases for dead passengers. But this model makes mistakes in 27% of cases for survived passengers. This algorithm prefers to predict death.

I also noticed one interesting fact. This 'accuracy' for every single class is exactly 'recall' in this sklearn report. I looked at that whole classification report. First, I thought that total accuracy is one simple average number from all recall values. But I checked this math again. That recall macro average is 0.79. This total accuracy is 0.8. This difference shows we calculate total accuracy differently.

After that, I did one cross-validation. I cut this data into 5 parts. I got very stable scores. In my Computer Vision class, neural network scores jump heavily. That happens because those networks use random starting weights. But this Logistic Regression model

always finds this exact same optimal minimum. This cross-validation proves this model learns real data rules.

Finally, I fixed one code error during this work. Pandas gave one warning about 'inplace=True'. I replaced that code line. This thing helps pandas work better. I feel good about this work.